



HK IMO Training 2022-2023
Problem Set 8
For 19 November 2022



Problem 29. Find all real numbers a and b such that whenever both quadratic equations $x^2 + ax + 1 = c$ and $x^2 + bx + 1 = d$ are solvable in real numbers (where c and d are real numbers), the quadratic equation $x^2 + (a + b)x + 1 = cd$ must be solvable in real numbers.

Problem 30. In $\triangle ABC$, $AB < AC$. Let M and N be the midpoints of AB and AC respectively. Let D be the intersection point of the line BC and the tangent at A to the circumcircle of $\triangle ABC$. The circumcircle of $\triangle ADN$ meets the line AB again at E . The lines BC and EN meet at F . Prove that DM is tangent to the circumcircle of $\triangle BFM$.

Problem 31. Find the smallest positive integer n that satisfies the following condition: It is possible to choose exactly 101 cells of an $n \times n$ table, such that there is a unique way to partition the table into rectangles (by cutting along the grid lines of the table) where each rectangle contains exactly one chosen cell.

Problem 32. Find all prime numbers p and q such that $3p^{q-1} + 1$ divides $11^p + 17^p$.